

2023 Ph H2 Q5

Section: Particles and Waves

Topic: Doppler Effect, LEDs, Photon Energy

(a)(i)

Calculate the frequency of the sound heard by the person as the police car approaches.

 **Given:**

- Frequency emitted by siren: $f_s = 700 \text{ Hz}$
- Speed of sound: $v = 340 \text{ m/s}$
- Speed of police car: $v_s = 34 \text{ m/s}$

 **Use Doppler equation (source approaching observer):**

$$f' = \frac{v}{v - v_s} \cdot f_s = \frac{340}{340 - 34} \cdot 700 = \frac{340}{306} \cdot 700 \approx 778.4 \text{ Hz}$$

Answer:

778 Hz

 **Marks: 3**

- 1 mark: correct formula
- 1 mark: correct substitution
- 1 mark: correct final answer

(a)(ii)

Explain why the pitch of the sound decreases as the car passes.

 **Before passing:**

- Source is **approaching** → wavelengths **shortened** → frequency **increases** → pitch rises

 **After passing:**

- Source is **receding** → wavelengths **lengthened** → frequency **decreases** → pitch falls

Answer:

Because the car is moving away, the observed frequency decreases,

 **Marks: 1**

(b)

State what is meant by monochromatic light.

 Mono = one Chromatic = colour → **one colour**
Light of a **single wavelength** or **frequency**

Answer:

Light with a single wavelength or frequency

 **Marks: 1**

(c)

The LED emits blue light with a wavelength of 460 nm. Calculate the energy of one photon.

 **Given:**

- $\lambda = 460 \times 10^{-9} \text{ m}$
- $h = 6.63 \times 10^{-34} \text{ J s}$
- $c = 3.00 \times 10^8 \text{ m/s}$

 **Use formula:**

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \cdot 3.00 \times 10^8}{460 \times 10^{-9}} \approx 4.32 \times 10^{-19} \text{ J}$$

Answer:

$4.3 \times 10^{-19} \text{ J}$

 **Marks: 3**

- 1 mark: correct formula
- 1 mark: correct substitution
- 1 mark: correct answer

 **Revision Tips:**

- Doppler effect: approaching → higher pitch; receding → lower pitch
- Monochromatic = single colour = one wavelength
- Photon energy: $E = \frac{hc}{\lambda}$
- Units must match! Use metres for wavelength, not nanometres